

AWS Machine Learning Engineer – Associate

FSx Deployment Options and Amazon Kinesis

week 3 – Part 3

Continuation of AWS storage concepts and introduction to real-time streaming data.

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- Scratch file systems are temporary and the data is not replicated. They are designed for short-term processing and high burst performance.
- Persistent file systems are for long-term processing. Data is replicated within the same Availability Zone and failed file servers can be replaced.
- Both can use an S3 bucket as an optional data repository.
- ML example: Scratch can fit temporary model-processing jobs when the original dataset is safely stored in S3; Persistent is better when the file-system data itself must remain available.

- FSx for NetApp ONTAP supports NFS, SMB, and iSCSI and helps move ONTAP or NAS workloads to AWS.
- ONTAP supports automatic storage growth/shrink, snapshots, replication, compression, deduplication, and point-in-time cloning.
- FSx for OpenZFS is a managed ZFS file system and supports NFS versions 3, 4, 4.1, and 4.2.
- OpenZFS can provide up to 1,000,000 IOPS with latency below 0.5 ms and also supports snapshots, compression, and cloning.

- Kinesis Data Streams collects and stores continuously arriving data in real time.
- Examples include click streams, IoT-device data, application events, metrics, and logs.
- Producers write records into the stream and consumers read and process those records.
- Consumers can include applications, AWS Lambda, Amazon Data Firehose, and Managed Service for Apache Flink.
- This is useful when data must be processed quickly instead of waiting for a complete batch file.

- Data can be retained for up to 365 days according to the course, which allows consumers to replay and reprocess retained records.
- Records remain in Kinesis until they expire rather than being manually deleted.
- The course lists a maximum record size of 10 MiB, while typical workloads often contain many smaller real-time records.
- Ordering is guaranteed for records that use the same partition ID.
- Kinesis supports KMS encryption at rest and HTTPS encryption in transit.
- KPL and KCL help developers create optimized producer and consumer applications.

- Provisioned mode requires the team to choose and manage the number of shards.
- Each shard provides 1 MB/s incoming throughput or 1,000 records per second, and 2 MB/s outgoing throughput.
- Provisioned capacity is manually scaled and pricing is based on provisioned shards.
- On-Demand mode removes manual capacity management and automatically scales based on observed throughput.
- The course lists default On-Demand capacity of 4 MB/s incoming or 4,000 records per second.
- Choose Provisioned for more direct capacity control and On-Demand when traffic is difficult to predict.

- Amazon Data Firehose is a fully managed service for delivering streaming data to destinations.
- It can deliver to Amazon S3, Amazon Redshift, Amazon OpenSearch Service, third-party services, and custom HTTP endpoints.
- It is serverless, automatically scales, and is considered near real time because records can be buffered by size or time.
- It supports CSV, JSON, Parquet, Avro, raw text, and binary data.
- Firehose can convert formats, compress records, and use AWS Lambda for custom transformations such as CSV to JSON.

- Kinesis Data Streams is used when applications need to collect, consume, process, retain, or replay a stream.
- It supports real-time processing, producer/consumer code, Provisioned or On-Demand capacity, and retention.
- Data Firehose focuses on loading streaming data into S3, Redshift, OpenSearch, third-party services, or HTTP destinations.
- Firehose is fully managed, automatically scales, and operates near real time.
- The course states that Firehose does not provide data storage or replay capability.
- Exam clue: need replay/custom consumers → Data Streams; need managed delivery to a destination → Firehose.

- For slow producers, check throughput limits and throttling and make sure partition keys distribute records evenly across shards.
- Large producers can batch records using KPL, PutRecords, or record aggregation.
- Use retries and exponential backoff for temporary service errors and investigate hot shards with enhanced monitoring.
- For slow consumers, check shard capacity, maxRecords, CPU/memory, and whether the consumer code itself is slow.
- Monitor `IteratorAgeMilliseconds` or `MillisBehindLatest` to see whether consumers are falling behind.
- For Lambda consumers, also check execution-role permissions, timeouts, concurrency limits, and throttling.

- FSx for Lustre Scratch is temporary and fast; Persistent is designed for longer-term replicated storage.
- FSx for NetApp ONTAP supports NFS, SMB, and iSCSI, while FSx for OpenZFS provides managed ZFS-compatible NFS storage.
- Kinesis Data Streams handles real-time streaming data and supports retention and replay.
- Provisioned Kinesis uses manually managed shards; On-Demand automatically manages capacity.
- Amazon Data Firehose provides managed near-real-time delivery and optional data transformation.
- Remember the exam distinction: Data Streams for consumers and replay; Firehose for managed delivery.
- Kinesis troubleshooting focuses on shard capacity, partition keys, throttling, retries, consumer lag, permissions, and application resources.